

Benchmark design governs reported skill in daily river water-temperature prediction

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Key points

- Confirmatory: seven-day skill falls from +0.250 against persistence to +0.038 against damped persistence, and every model collapses alike.
- Confirmatory: a hybrid reporting +0.218 against persistence is indistinguishable from the damped baseline; the reference set the sign.
- Confirmatory: keys a model may decline are 17-25% harder than the keys it keeps, across six models from persistence to deep learning.

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Abstract

Daily river water temperature is strongly persistent and seasonal, so reported machine-learning skill depends on what a model is compared against and on which days it is scored. On one frozen panel of 120 U.S. gauges across 15 hydrologic regions we hold the models and data fixed and vary the evaluation design, scoring every arm on one common registry of station/date/horizon keys with all preprocessing fitted strictly backwards in time, on a pre-specified 2021-2023 held-out window of 116 reportable stations. Two channels are quantified. The reference: seven-day station-median skill falls from +0.250 against persistence to +0.038 against damped persistence, a property of the panel and not of one model — all six compared models lie between +0.218 and +0.250 against persistence and between -0.011 and +0.038 against damped persistence, so an air2stream-style hybrid appearing to capture a fifth of the seven-day error budget is indistinguishable from the baseline. The key set: one model failed on 0.75% of keys, and six witness models all score 17-25% worse on the keys it declined than on those it kept, so a complete-case convention would have scored it on a measurably easier subset. A gradient-boosted tree with site identity is most accurate at one and three days. Post-outcome descriptive analyses then find that withholding a gauge's own recent water temperature costs 1.3-1.7 °C, two orders of magnitude more than the gap between the two estimator implementations compared — hypotheses this cohort generated, not findings it confirmed, and each an input ablation at gauged sites.

Plain Language Summary

River temperature changes slowly, so a forecast repeating yesterday's reading is already fairly accurate, and nudging it toward the season's usual value is better still. That makes the comparison a model is judged against decisive. We tested predictions at 120 U.S. gauges under one demanding evaluation: every model scored on exactly the same days and sites, none allowed to see anything dated after its issue time. Judged against the naive forecast all six methods looked about equally good and all looked strong; judged against the slightly smarter one they separated, and one established method proved no better than the simple rule it had appeared to beat by a fifth. The days a method is allowed to skip, in studies that let it, are also markedly harder than the days it keeps, so skipping flatters it. Separately and more tentatively, what a model was *given* mattered more than which method it was: denying it the recent readings from the gauge it predicts cost 1.3 to 1.7 °C, against under 0.01 °C for switching model implementations. Much of what is reported as a modelling advance is, on this evidence, a statement about the comparison and the days scored.

Keywords: river water temperature; benchmark design; strong baselines; spatial holdout; comparative evaluation; information value.

1 Introduction

Stream thermal regimes set dissolved-oxygen saturation, metabolic rates, life-stage timing, and habitat suitability, and respond to atmospheric forcing, discharge, groundwater exchange, shading, channel geometry, and regulation (Caissie, 2006). Daily records from national monitoring networks have made the variable one of the most heavily modeled targets in applied hydrological machine learning, and the prevailing account is that deep sequence models have substantially advanced it: non-linear air–water regressions (Mohseni et al., 1998) and hybrid air-temperature and discharge formulations (Toffolon & Piccolroaz, 2015) were succeeded by recurrent, multi-task, and physics-guided river-network architectures reporting large error reductions (Rahmani et al. (2021); Sadler et al. (2022); Barclay et al. (2023)) and by graph architectures making spatial dependence explicit (Topp et al., 2023).

Those gains are heterogeneous and the heterogeneity does not organize itself by architecture. Published accuracies are not comparable because reference models, key sets, and splitting conventions differ (Corona & Hogue, 2025), and where families *are* compared under a common protocol the spread is small: neither a comparison of statistical and machine-learning water-temperature models including an air2stream formulation (Feigl et al., 2021) nor one across recurrent, convolutional, and transformer architectures (Liu et al., 2025) finds an ordering that would explain the published dispersion. That a score is a statement about its benchmark is itself uncontested, from the differential split-sample test (Kleměš, 1986) through explicit upper and lower benchmarks (Seibert et al., 2018) to benchmarking as the centre of what machine learning can tell us about a hydrological system (Nearing et al., 2021). What is missing for daily river temperature is the measurement.

This paper measures two channels of it on identical keys. The first is the reference: daily mean water temperature has enough thermal inertia that relaxing the last observation toward a seasonal climatology already removes most of the error a learned model can remove, so the denominator of a skill score is a modelling choice with the size of a result. The second is the key set: models are routinely scored on model-specific complete-case sets, so a model can gain apparent accuracy by declining to predict on hard days. A third channel, preprocessing fitted on windows that include the evaluated interval (Arsenault et al., 2018), is controlled here by construction but not quantified, and a fourth — random held-site splits, where a held gauge's neighbours remain in training (Kratzert et al., 2019) — turns out to be below this design's resolution (Section 4.4). None of the four is visible in a reported RMSE.

Two boundaries follow, and recent work sits beyond both. Every model here consumes issue-time information only and is scored on retrospective gridded meteorology, not on what a forecaster could have run (Zwart, Oliver, et al. (2023); Padrón et al. (2025)); and every confirmatory arm retains the target gauge's own observed history, so nothing here speaks to reaches that were never instrumented,

now addressed at CONUS scale by frameworks built for that case (Bakar Siddik et al. (2026); Philippus et al. (2026)) and in far sparser networks (Chang et al., 2025).

The panel is 657,480 site-days from 120 stable USGS site numbers over 2006–2020, in 34 states and 15 two-digit hydrologic unit (HUC2) regions; every model is tuned on data through 2020 and scored on one common set of station/date/horizon keys at 1-, 3- and 7-day leads, with a held-out 2021–2023 window. The cohort is availability-selected rather than drawn, bounding what it can generalize to (Sections 3.4, 6). ThermoRoute, whose skill is decomposed, is described in Section 3.1; its architecture is the object under test, not the contribution.

2 Data and design

2.1 The frozen panel and the cohort it defines

The development record is a single derived panel bound by a manifest to a station registry (SI01). Each of 120 sites carries one row per calendar day from 2006-01-01 through 2020-12-31, giving 657,480 rows, missing observations being explicit nulls rather than absent rows. The models consume seven raw variables — water temperature, streamflow, air temperature, precipitation, a relative-humidity proxy, daylight-period mean shortwave radiation and wind speed — from USGS NWIS daily values (United States Geological Survey, 1994), Daymet V4 (Thornton et al., 2022) and gridMET (Abatzoglou, 2013).

The cohort was not sampled at random and nothing treats it as though it were: selection rejected 1,345 of 1,465 candidates, mostly for lacking joint water-temperature and discharge availability, and 2019–2020 participated in its construction. Water temperature is absent on 15.8% of rows and discharge on 2.8%; 38 retained stations share a hydrologic unit code with another and 19 lie within 10 km of another, so station-level independence is not tenable.

2.2 Temporal roles and the common key set

Temporal roles are fixed before any model is fitted, and a training sample is admitted only if its issue date and all target dates fall in the same partition: training 2006–2015 (438,240 rows, 341,646 with observed water temperature), validation 2016–2017, calibration 2018, development evaluation 2019–2020, and the held-out test 2021–2023 (counts in SI01). Models are tuned exclusively on data through 2020. The held-out window was opened once for the pre-specified Section 4.3 evaluation and then *reused* for the explicitly post-outcome analyses of Sections 4.4–4.6; it never informed tuning.

The key registry is a data rule, not a model outcome. A key is admissible when its issue and target dates fall inside the window, issue-date water temperature is finite and observed, a 32-day history can be built, and target-date

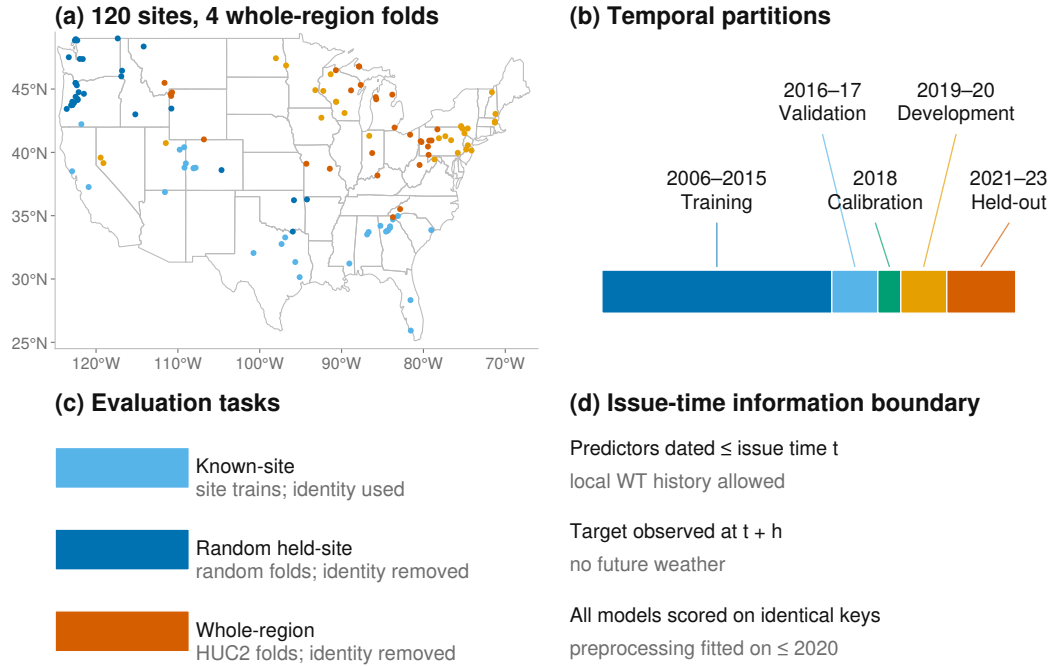


Figure 1. Study sites and evaluation design. (a) The 120 U.S. gauges coloured by the four deterministic whole-HUC2-region folds. (b) Temporal partitions. (c) The three evaluation tasks: known-site forecasting, random held-site transfer, and whole-region gauged transfer. (d) The issue-time information boundary — every model scored on the identical common forecast-key registry, no model-specific complete-case set permitted.

water temperature is finite and observed under the inclusive qualifier policy, evaluated independently at each lead. No condition mentions a model, and every declared model must produce a row on every admissible key: failing to predict is a failure, not an exclusion. Section 4.1 measures what that rule is worth. The development registry holds 249,072 keys, 83,024 per lead, and never carries a held-out claim.

2.3 Evaluation-period inputs and the information boundary

The test interval is 2021-01-01 through 2023-12-31 for the same cohort, with meteorology taken at each station coordinate rather than aggregated over upstream catchments — a real limitation for large basins (Section 6). The primary information set uses provider values dated no later than each issue date and consumes no horizon-specific forecast: a date-indexed retrospective hindcast, not a re-execution of what a forecaster could have run, because the as-issued provisional vintage of a gridded product cannot be reconstructed afterwards.

Water temperature (NWIS 00010, °C), discharge (00060) and gage height (00065) come from NWIS as the provider's 00003 daily mean on site-local dates. **Discharge is served in $\text{ft}^3 \text{s}^{-1}$ and consumed in those units:** it enters the models only after per-station standardization, so the unit affects no reported quantity, but the panel column is not $\text{m}^3 \text{s}^{-1}$. Values are retained irrespective of approval

qualifier; 2,059 of 657,480 rows carry negative discharge, a real measurement at tidal and regulated reaches, and observed water temperature spans -0.90 to 34.30 °C. The qualifier ledger and the provider-specific day definitions are in SI12. Test-window daily means are outcomes only: no model-, feature-, threshold-, calibration- or station-selection code reads them.

3 Methods

3.1 The predictor under test

ThermoRoute predicts at issue time t , station i and lead h by adding a bounded learned residual to a damped-persistence anchor,

$$(1) A_{i,t+h} = c_{i,t+h} + \phi_i^h (y_{i,t} - c_{i,t})$$

where c is a seasonal climatology and ϕ_i a per-station decay fitted by no-intercept least squares on training-period consecutive-day anomaly pairs. The same object, fitted identically, is the damped-persistence reference. The residual comes from a strictly left-looking temporal convolutional encoder over a 14-lag window, with a horizon-conditioned router, a three-expert mixture and a ± 1 °C algebraic bound on the deviation from the anchor (Figure 2); one model serves all three leads, 38,505 trainable parameters, five seeds averaged (Text S3). The architecture is the object under test rather than the contribution: Section 4.2 shows an information-matched plain causal network reproduces it to within 0.023 °C.

3.2 The reference set

The reference set is what a reported gain is a gain over, and each member removes one alternative explanation for apparent skill. **Persistence** ($\hat{y}_{-\{t+h\}} = y_{-t}$) quantifies raw series memory; **damped persistence**, equation (1) fitted on the training period, is the primary reference; **climatology** isolates the seasonal cycle. **LightGBM** (Ke et al., 2017) is the principal learned reference, run globally with site identity and per station; a **global LSTM** with a station embedding represents the deep sequence family (Rahmani et al. (2021); Zwart, Diaz, et al. (2023)); a **plain causal temporal convolutional network** is the information-matched deep reference, with the same outcome-free inputs, a parameter count matched to within 2% (38,346 against 38,505) and an unrestricted residual around the same anchor; and an **air2stream-style hybrid** (Toffolon and Piccolroaz, 2015) is fitted per station on the same keys (SI07). Tuning budgets are documented but *not* equalized across classes, so any statement about LightGBM is scoped to this four-candidate procedure and feature schema rather than to gradient boosting in general.

Same information, same keys — the only fork is the regression target

Protocol schematic; no results shown. Tuning is not equalised; the optional F3 forcing axis is not drawn.

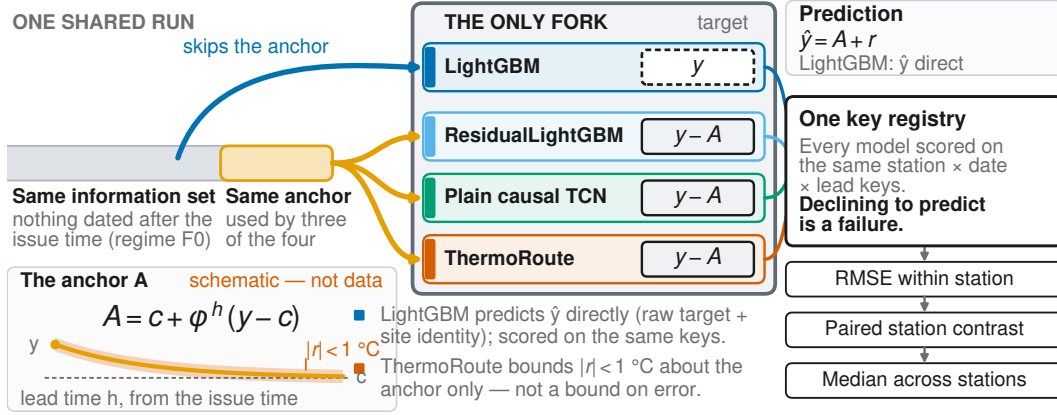


Figure 2. Everything the compared models share, and the one thing they do not.

A single run supplies every model the same information set — nothing dated after the issue time (regime F0) — and the same damped-persistence anchor A (inset), which three of the four predict a residual around while LightGBM predicts the raw target y directly with site identity. That target is the only fork. All models are then scored on one registry of station $\times$ date $\times$ lead keys, on which declining to predict counts as a failure rather than an exclusion, and reduced by the same station-first path: RMSE within station, paired contrast, median across stations. The ± 1 °C bound constrains the residual about the anchor, not the forecast error. The figure is structural and carries no evidence from either period; tuning is not equalised and the optional F3 forcing axis is not drawn.

3.3 Leakage control and spatial partitions

Leakage control is a set of mechanisms checkable against the code, not a statement of intent: every predictor carries a date no later than the issue date, the encoder is strictly left-looking, standardization constants, q90 event thresholds (2006–2015) and the 2018 conformal offsets and Platt calibrators are each fitted strictly before the interval they apply to, and a fresh interpreter replays every trained member rather than trusting self-consistent prediction files. The mask covers the primary predictor path; the issue-date auxiliary quantities behind the relaxation proposal and the regime gate share it rather than carrying one of their own, so invariance to fill values is enforced for the predictors and not proved for those two paths (SI02).

Random held-site splits can be optimistic, because a held gauge's neighbours remain in training and the model reaches its thermal regime through correlated forcing and shared preprocessing. We therefore compare four balanced random folds against four leave-HUC2-region folds of [30, 30, 31, 29] stations, at a mean 289 km from the nearest training gauge (protocol v1; SI11). Held sites still supply their own water temperature through the issue date, so this is **gauged** transfer to a region whose gauges were not fitted, not prediction at a site with no record. Split-conformalized intervals are calibrated on 2018 only and reported as a marginal coverage diagnostic (SI08).

3.4 Estimand, metrics, and comparison set

Two differently signed quantities are reported and never combined: the paired effect **(2)** $\Delta\text{RMSE} = \text{RMSE}(\text{cand}) - \text{RMSE}(\text{ref})$ in $^{\circ}\text{C}$, where **negative** favors the candidate, and the skill score **(3)** $\text{skill} = 1 - \text{RMSE}(\text{cand})/\text{RMSE}(\text{ref})$, dimensionless, where **positive** does. Every value in Section 4 is labeled with which it is. The sampling unit is the station: unweighted RMSE is computed on the common daily keys per reportable station and the primary effect is the unweighted median across stations, a station/lead cell being reportable only with at least 100 valid paired targets.

The comparison set is five rows and they are not the same kind of test: three compare ThermoRoute with damped persistence at 1, 3 and 7 days as *superiority* tests at a 0.00 $^{\circ}\text{C}$ margin, two compare it with LightGBM at 3 and 7 days as *non-inferiority* tests at a +0.05 $^{\circ}\text{C}$ margin. That margin is a numerical ceiling on allowable degradation fixed before the outcomes were opened; it is not derived from sensor precision, biological response or regulation, and nothing here is written as an equivalence or parity statement. Two uncertainty procedures accompany each row, both clustered at HUC2: a one-sided p from exact enumeration of all 2^K whole-cluster sign vectors against that row's own margin, and a 10,000-draw cluster bootstrap percentile interval, Holm-adjusted (Holm, 1979) over the five. A row is supported only when both agree. The cohort came from an availability filter rather than probability sampling of regions, so exchangeability is not established and the clustered intervals are approximate sensitivities, not decision evidence.

4 Results

Three evidence grades are used and they are not interchangeable. Sections 4.1 and 4.3 are *confirmatory*: suite, family, estimand, margins and decision rules were fixed on data through 2020 and the window opened only to score them. Section 4.2 is a *development diagnostic* on the 2019–2020 partition, which participated in cohort construction. Sections 4.4–4.6 are *post-outcome descriptive*, specified after the window had been opened; they carry that label wherever they are quoted, and SI20 maps each to its grade, artifact and specification date. Every value below is an unweighted station median over reportable stations on the relevant common key registry, in $^{\circ}\text{C}$.

4.1 Two channels of benchmark design, measured on identical keys

The reference collapse is a property of the panel, not of one model. On identical keys the same fixed model reports a seven-day median station skill of +0.251 against persistence and +0.038 against damped persistence, a factor of 6.6 between two numbers differing only in the denominator of equation (3). That would be a fact about ThermoRoute if it stopped there. It does not: every model in the

suite behaves the same way (Table 1). All six lie between +0.218 and +0.250 against persistence — a band narrow enough that a reader comparing two papers quoting persistence-relative skill is comparing numbers that never separated the models — and spread from −0.011 to +0.038 against damped persistence, half again as wide.

One model changes sign. The air2stream-style hybrid reports +0.218 [0.167, 0.233] against persistence and −0.011 [−0.031, +0.007] against damped persistence. The same fitted object, on the same days, either captures a fifth of the seven-day error budget or is indistinguishable from the baseline, according only to which reference sits in the denominator. Rebuilding the anchor under seven pre-declared one-factor variants moves ThermoRoute's seven-day skill from +0.025 to +0.063 (`outputs/final/anchor_sensitivity.parquet`); the better the anchor, the less any learned model adds.

The key set is the second channel, and this cohort measures it. Fourteen of fifteen scored models predict on all 358,807 held-out keys; the hybrid produces 356,131, failing on 2,676 (0.75%) concentrated in nine of 118 sites. Under the common-key rule that is a failure and everyone is scored on the intersection. Under the complete-case convention much of the literature uses it would be an exclusion, and the hybrid would be scored on a subset selected by its own ability to handle it. Whether that subset is easier needs no counterfactual model, because fourteen models did predict those keys: **every one of six witnesses spanning persistence to deep learning scores 17–25% worse on the declined keys than on the rest** (median ratio 1.21; ThermoRoute 1.560 against 1.300 °C, damped persistence 1.720 against 1.396 °C; `outputs/final/key_set_counterfactual_v1/`). The days a model drops, when it is allowed to drop days, are materially harder than the days it keeps. That is the size of the advantage the common-key registry removes by construction.

Table 1 — every model, its accuracy, and what it reports against each reference. Station-median RMSE in °C at each lead and median station skill (equation 3) at seven days, all on the identical held-out registry, 116 reportable stations, with whole-HUC2 cluster-bootstrap intervals. Reading down a column compares models; reading across a row shows what the reference alone does to one fitted object. The three reference models carry no skill against themselves. MAE and bias are the same rows in SI07.

Model	RMSE 1 d	RMSE 3 d	RMSE 7 d	Skill 7 d vs. persistence	Skill 7 d vs. damped
Persistence	0.813	1.638	2.202	—	—
DampedPersistence	0.789	1.454	1.773	—	—
Climatology	1.899	1.902	1.903	—	—
LightGBM	0.589	1.304	1.735	+0.248 [+0.230, +0.263]	+0.030 [+0.027, +0.037]
LSTM	0.663	1.358	1.712	+0.244 [+0.225, +0.256]	+0.028 [+0.027, +0.034]
PlainMLP-7var	0.690	1.374	1.720	+0.243 [+0.220, +0.255]	+0.023 [+0.020, +0.032]
PlainCausalTCN-7var	0.646	1.330	1.710	+0.246 [+0.223, +0.258]	+0.028 [+0.024, +0.034]
Air2stream	0.719	1.459	1.825	+0.218 [+0.167, +0.233]	−0.011 [−0.031, +0.007]
ThermoRoute	0.640	1.337	1.694	+0.250 [+0.235, +0.268]	+0.038 [+0.033, +0.046]

4.2 A tree is the most accurate model, and the architecture is reproducible

On the development window the tree has the lowest station-median RMSE at all three leads (0.578, 1.280 and 1.649 °C against ThermoRoute's 0.631, 1.291 and 1.657 °C), with paired differences of +0.046, +0.008 and −0.005 °C: on this panel the ranking does not favor architectural constraint, which is the result this study is built to detect. The architecture's distinguishing property is instead the ± 1 °C algebraic bound, which held on 100.00% of audited rows as a guarantee of the construction rather than a fitted outcome. The deep predictor is also reproducible from a much simpler object — an information-matched plain causal network tracks it to within 0.023 °C at every lead, and of the one-factor deletions only removing the temporal encoder moves the 1-day RMSE materially (SI09).

4.3 Held-out 2021–2023 evaluation

The held-out evaluation applies the Section 3 suite and comparison set, fixed on data through 2020, to 2021–2023 on the common registry ($n = 116$; per-model

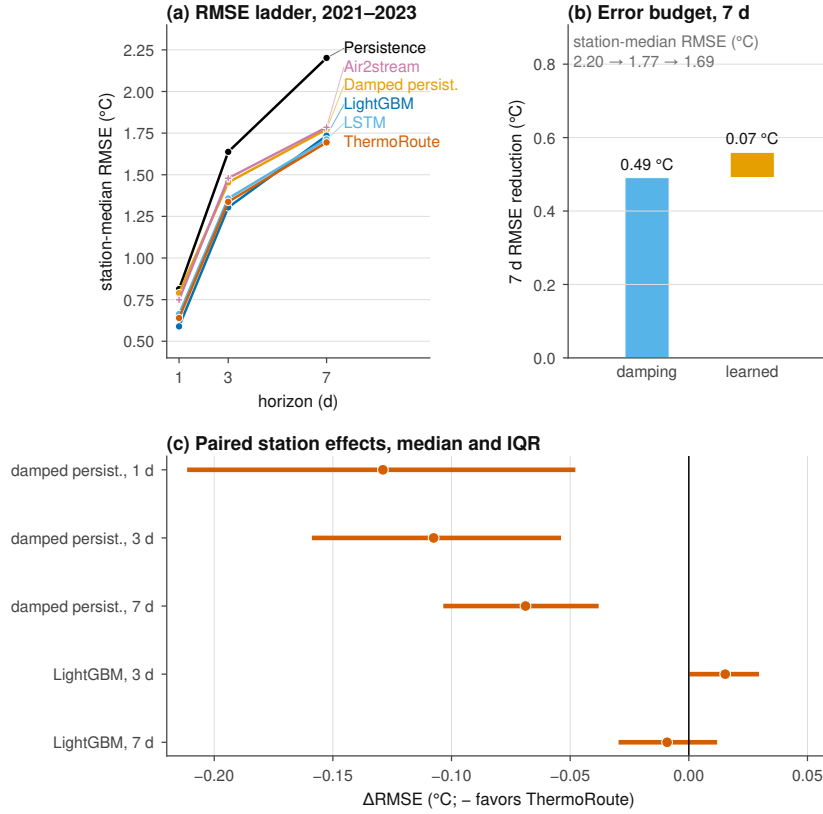


Figure 3. Decomposition of reported skill on the held-out 2021–2023 window (116 reportable stations). (a) Station-median RMSE by lead, every compared model. (b) The seven-day error budget: 0.49 °C of the reduction comes from damping the anchor and 0.07 °C from the learned residual. (c) Paired station effects for the five sealed tests, median and interquartile range; negative values favor ThermoRoute. Panel (c) shows the dispersion of the paired effects, not the narrower cluster-bootstrap intervals of Table 2.

metrics in SI07). Station-median skill is +0.250 against persistence and +0.038 against damped persistence, the tree with site identity leads at one and three days (0.589 and 1.304 °C), and at seven days ThermoRoute reaches 1.694 °C against the tree's 1.735 °C. Against the information-matched plain convolutional network the paired station median is −0.015 °C at seven days in ThermoRoute's favour, a contrast outside the sealed family carrying no decision.

Each test is read at the margin it was sealed at (Section 3.4). All five rows meet their sealed decision rule. Two readings are worth stating: row 4 meets a non-inferiority margin at a lead where the tree is the better model (+0.015 °C, interval excluding zero from above), and at seven days the median of −0.009 °C has an interval covering zero, so that row establishes the ceiling rather than a win. The seal is repository-internal Git and SHA-256 evidence and its scope is set out in SI20.

Table 2 — the five sealed tests, each at its registered margin. ThermoRoute is the candidate and 116 stations are reportable throughout. Station-level ΔRMSE in °C, negative favors ThermoRoute; whole-HUC2 cluster-bootstrap inter-

vals; sign-flip p computed against the row's own margin, Holm-adjusted (unadjusted $p \leq 6.1 \times 10^{-5}$ throughout, the floor for 15 clusters). Rows 4 and 5 were registered as non-inferiority tests, so a margin of zero would misreport them. "Meets margin" is the literal reading and not a claim that the candidate won: row 4 meets its margin at a lead where the tree is the better model (`outputs/final/sealed_confirmatory_family_v1/`).

#	Reference	Lead	Margin (°C)	Δ RMSE (°C)	95% CI	Win rate	Holm p	Sealed decision
1	DampedPersistence	1 d	+0.00	-0.129	[-0.199, -0.076]	0.90	1.5e-04	Meets margin
2	DampedPersistence	3 d	+0.00	-0.108	[-0.143, -0.074]	0.91	1.5e-04	Meets margin
3	DampedPersistence	7 d	+0.00	-0.069	[-0.084, -0.057]	0.95	1.8e-04	Meets margin
4	LightGBM	3 d	+0.05	0.015	[0.010, 0.025]	0.24	1.8e-04	Meets margin
5	LightGBM	7 d	+0.05	-0.009	[-0.017, 0.003]	0.59	1.8e-04	Meets margin

4.4 Spatial transfer on 2021–2023

The development-period whole-region finding (SI19) is re-tested on the independent window under four leave-HUC2-region folds and four balanced random folds over five split seeds (protocol v1; SI11, Figure S11), moving the median nearest-training-gauge distance from 60 to 263 km. The region-minus-random paired penalty for the raw-target tree is +0.006, +0.009 and +0.007 °C.

That penalty is the size of its own resampling noise — the median per-site standard deviation across five seeds is 0.004–0.007 °C — and four folds give four independent regional observations, so the geometry contrast sits **below this design's resolution** and no geometry effect is claimed in either direction. The numbers are the local-adaptation condition; the pooled-adaptation arm is reported separately in SI11, where a preprocessing defect confines it to a sensitivity.

4.5 Hydrologic conditions governing incremental skill

The remaining gain concentrates in specific states. Station thermal memory — the anomaly half-life of the train-fitted anchor, not an e-folding time — has a median of 6.9 days (IQR 5.0–11.9). At seven days the median memory gain is 0.49 °C against a learned gain of 0.07 °C (Figure 3b), the median per-station fraction of the reduction delivered by seasonal memory is 0.875, and the correlation between log half-life and learned gain is -0.40 at one day: longer-memory stations leave less to add. With thresholds fitted on 2006–2015 only, the seven-day median

paired ΔRMSE over damped persistence is -0.069°C on all keys and largest under low thermal anomaly (-0.229°C) and rapid recent cooling (-0.170°C), reaching -0.088°C under rapid recent warming, with no issue-time state reversing the ordering. The sixteen-stratum table, the outcome-conditioned strata that a manager cannot identify at issue time, and the reportability and multiplicity caveats that keep them descriptive are in SI11.

4.6 What the information is worth, and where the estimator starts to matter

Sections 4.4 and 4.5 leave one question open: the residual learned gain is small, but is that a limit of the models or of what they were given? Four crossed axes answer it, and **every result here is post-outcome and descriptive**. **L** withholds the target site's own observations, **F** substitutes realized meteorology over $t+1 \dots t+h$ as an oracle bound, **G** is the holdout geometry and **A** swaps the tree for the information-matched network. Prohibited inputs are dropped rather than filled, all 24 mask-invariance proofs return exactly zero, and a negative control re-admitting one water-temperature lag is caught with a 147°C shift. Levels, estimands, sealing and per-cell tables are in SI20.

Local thermal state dominates. Withholding it costs $1.3\text{--}1.7^\circ\text{C}$ at every lead and at all 116 stations — two to thirteen times realized future meteorology and roughly seventy times the architecture effect of Section 4.2 — and almost all of that is the recent sequence rather than the long record ($0.01\text{--}0.16^\circ\text{C}$ to pool a station's climatology and damped rate, $1.17\text{--}1.71^\circ\text{C}$ to remove its recent readings). Neither ablation tests prediction at a site that was never instrumented: a station with no thermal record has no target to score against.

Realized future meteorology is worth 0.13°C at one day and $0.54\text{--}0.63^\circ\text{C}$ at three and seven, three to four times each cell's minimum detectable effect, and is event-scale weather information rather than seasonal phase: a placebo whose rules were fixed before its outcome existed retains $0.0\text{--}8.8\%$ of the value, and it is almost entirely air temperature.

Table 3 — value of realized future meteorology. Station-first forcing value $V_F = \text{median}_i[\text{RMSE}_i(F0) - \text{RMSE}_i(F3)]$ in $^\circ\text{C}$; CI is a 10,000-draw whole-HUC2 cluster bootstrap and MDE the smallest effect this cohort's cluster structure resolves at $\alpha = 0.05$.

Model	Lead	RMSE F0	RMSE F3	V_F	95% CI	MDE
LightGBM	1 d	0.614	0.451	0.130	[0.081, 0.189]	0.031
LightGBM	3 d	1.330	0.765	0.542	[0.415, 0.721]	0.152
LightGBM	7 d	1.742	1.063	0.627	[0.423, 0.803]	0.172
ResidualLightGBM	1 d	0.606	0.446	0.125	[0.084, 0.183]	0.028
ResidualLightGBM	3 d	1.327	0.752	0.578	[0.406, 0.704]	0.160
ResidualLightGBM	7 d	1.722	1.087	0.605	[0.445, 0.795]	0.151

Crossed with L its marginal value is *lower* once local observations are withheld, but only at short leads (Table 4) — a statistical interaction within the evaluated design, not a general property of the two information sources.

Table 4 — value of realized future meteorology by information level. Station-first paired values in °C under whole-region holdout, 116 reportable stations, 10,000-draw whole-HUC2 cluster bootstrap.

Forcing value	1 d	3 d	7 d
At L0, gauged	0.116 [0.059, 0.174]	0.474 [0.299, 0.693]	0.580 [0.366, 0.779]
At L2, obs. withheld	0.038 [0.020, 0.072]	0.331 [0.249, 0.496]	0.596 [0.431, 0.804]
Interaction, L2 – L0	−0.076 [−0.098, −0.056]	−0.147 [−0.193, −0.079]	+0.016 [−0.035, +0.099]

The estimator matters only where the information does not. At L0 the tree and the network differ by at most 0.009 °C at any lead against station-median errors of 0.53–1.74 °C, under both geometries — two fitted objects, not two model families, since their tuning budgets were documented rather than equalized (Section 3.2) and no graph, recurrent or differentiable-hybrid model is represented. Remove the local gauge and the tree wins by 0.02–0.19 °C; given the oracle the sign reverses, to −0.096 [−0.177, −0.038] at seven days, so which estimator wins is itself conditional on the information regime. The two triple differences fall below their cells' minimum detectable effect and are reported as unresolved.

5 Discussion

5.1 Why these numbers are smaller than published gains, and what transfers

The difference is design, not model quality, and Section 4.1 puts numbers on two of the channels. A study quoting skill against persistence or climatology reports something comparable to our +0.203 to +0.251 column rather than our +0.038 to +0.168 column — up to a factor of 6.6 for the same fitted weights, and enough to move a hybrid across zero — so we suggest damped persistence as the minimum reference for this variable, alongside the naive one rather than instead of it. A study

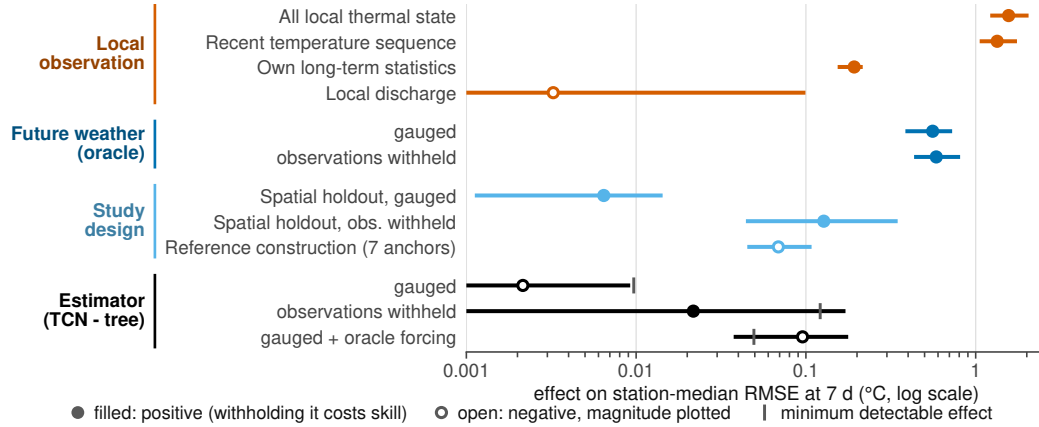


Figure 4. What each axis of the study is worth, at seven days. Station-first paired effects in °C on a log axis, damped-residual tree throughout; open markers are negative effects plotted at their magnitude, bars that reach the 0.001 floor are intervals spanning zero, and the tick marks the minimum detectable effect where one is defined. Within *Future weather (oracle)* and *Estimator* the row label gives only what distinguishes it — “gauged” is L0 and “observations withheld” is L2 — and the bracket carries the rest, so *Estimator* rows are the plain causal network minus the tree. Local observation, future weather, study design and the estimator span three orders of magnitude on one continuous axis; the axes are separate conditional designs and are compared, never summed. Every row is post-outcome and descriptive except *reference construction*, whose bar is also the one interval that is not a cluster bootstrap: it is the range across the seven predeclared anchor variants of Section 4.1, a spread over specifications. It is on this axis because every other row is conditional on one anchor while the anchor is itself a fitted object, and it says how much of the figure is a property of the reference rather than of the river.

scoring each model on its own complete-case set gives the model with the most gaps the easiest days: here the declined keys ran 17–25% harder for every model that attempted them.

The evaluation design is nonetheless stronger than its inferential reach. Clustered inference rests on resampling whole spatial units and needs enough of them, of comparable size, for the asymptotics to hold; fifteen HUC2 groups with an inverse-Herfindahl effective count of 9.54 and a largest group holding 21.7% of stations do not meet that bar, and the cohort was never exchangeable across regions. Usual practice would report a station-level interval, and nothing in such a paper would tell a reader the effective cluster count is under ten — the difference is disclosure rather than the data or the model.

Why our estimator gap is smaller than published comparisons, and why an operational study would rank differently. Two recent results are the right external checks on ours and neither contradicts it. Feigl et al. (2021) compare six model classes on Austrian catchments and report a median RMSE spread between them of order 0.08 °C, where our L0 contrast between two implementations is 0.009 °C. The two numbers measure different things: theirs spans a wider set of model families each tuned in its own way, ours holds the information set, the regres-

sion target and the key registry fixed and varies only the estimator, so ours is the narrower quantity and should be smaller. Read together they say the same thing — the estimator is not where the difficulty lies — with ours putting a floor under how small the gap becomes once the controls are matched.

Padrón et al. (2025) forecast stream temperature at extended range with archived weather and find model choice mattering more than we do, and the reason is visible in our own axes. Their setting supplies neither of the two conditions under which our estimator contrast vanishes: their forcing is a real forecast rather than issue-time-only information, and their harder cases withhold the target site's history. Both of those are cells where our own estimator penalty becomes non-zero and, given the oracle, changes sign. An operational study should therefore expect a different ranking from ours, and that is a prediction this design makes rather than a disagreement.

What this implies for reporting. Three things would make published river temperature results comparable at negligible cost: quote skill against a damped anchor alongside the naive one, since the pair costs one extra column and is what separates the models; state how many keys each model actually scored, since a model that declines the hard days is credited for an easier subset; and declare whether preprocessing statistics were fitted before or across the evaluated interval. None requires new computation, and the first two are what this study measures.

The study is narrow by construction: a common-key clustered benchmark of point predictors at gauged sites, not an attempt to replace process-based thermal models, river-network graph models or differentiable hybrid formulations (Jia et al. (2021); Rahmani et al. (2023); Zwart, Diaz, et al. (2023)). What is portable is not the architecture but three controls: one key registry across models, statistics fitted strictly backwards in time, and partitions by whole hydrologic region.

6 Limitations

Scope. The estimand is a fixed availability-selected cohort of gauged sites, and the clustered intervals are approximate sensitivities on it (Section 5.2). Section 4.6 withholds a station's own record from the model's inputs, which is a different question from prediction at a location that was never instrumented: the cohort, the climatology and the evaluation keys all come from gauged sites, and this study does not address the ungauged case. Throughout, "causal" describes time ordering only.

Cohort and measurement. Requiring joint water-temperature and discharge records buys a discharge channel worth $+0.042$ °C at 1 day (SI19) at the cost of 1,345 of 1,465 candidates, which enriches the panel for long, complete records and bounds what it generalizes to. Meteorology is taken at station coordinates rather than integrated over upstream catchments. The air2stream-style reference is an em-

pirical variant of the published formulation (Toffolon and Piccolroaz, 2015) rather than the official code, and is used as a reference rather than as evidence about the process model.

Reference and thresholds. The damped anchor is a fitted object and the seven-day headline moves by a factor of 2.5 across defensible constructions of it (Section 4.1), so every number against damped persistence is conditional on the anchor of Section 3.1, primary because it was fixed before the window opened. The ± 1 °C bound, the event quantiles and the 100-key rule are development-selected, and q90 is a statistical tail diagnostic (SI19).

7 Conclusions

We evaluated daily river water-temperature prediction at 120 stable U.S. gauges across 15 hydrologic regions on one common set of station/date/horizon keys, all statistics fitted strictly backwards in time, on a held-out 2021–2023 window.

Two channels of benchmark design were quantified, and both are properties of the panel rather than of any one model. The reference governs the reported gain: seven-day station-median skill falls from +0.250 against persistence to +0.038 against damped persistence, every model in the suite collapses the same way, and an air2stream-style hybrid that reports +0.218 against persistence is indistinguishable from the damped baseline. The key set governs which days a model is asked about: the keys one model declined are 17–25% harder than the keys it kept, unanimously across six witnesses, so a complete-case convention would have scored it on an easier subset than the registry it was drawn from.

Model ranking does not favor architectural constraint — a gradient-boosted tree with site identity is most accurate at one and three days — and the spatial-partition question is unresolvable in the regime it was posed in. A post-outcome descriptive analysis suggests the information a model is given dominates the estimator by two orders of magnitude, but that is a hypothesis this cohort generated rather than one it confirmed.

A published skill score is not interpretable without its reference model, its key set and its information set, and every comparison here is descriptive for this fixed, availability-selected cohort.

Open Research Statement

Data. The derived panel, the station registry, their manifests and the test-window acquisition record are deposited as one versioned dataset at [DATA DOI TO BE MINTED] under [DATA LICENCE TO BE ASSIGNED], pending the byte-level rights review, since a derived product does not inherit the most permissive of its upstream terms; where redistribution terms are unresolved the deposit carries the request specification and a SHA-256 manifest in place of the bytes (SI16). Primary sources

are USGS NWIS (United States Geological Survey, 1994), Daymet V4 R1 (Thornton et al., 2022) and gridMET (Abatzoglou, 2013), each under its own terms.

Software and reproduction. The analysis software, per-stage manifests and Python 3.12 dependency lock with package hashes are archived at [SOFTWARE DOI TO BE MINTED]; the MIT licence covers neither the observational data nor the archived provider responses. Sections 4.1–4.2 reproduce from the archived panel, bundles and pinned environment and Section 4.3 from the acquisition record, with bit-level equality expected for the trees and a documented tolerance for the neural members (SI15). Independent reproduction on a third-party host is planned and has not yet been performed.

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Supporting Information

SI01 cohort, registry and partition counts; SI02 issue-time boundary; SI03 equations and units; SI04 protocol, chronology and decision log; SI05–SI06 the comparison set; SI07 all-model scores with MAE and bias; SI08 probability and reliability schemas; SI09 architecture and information-matched controls with seed and budget provenance; SI10 temporal coverage; SI11 the spatial folds, cluster geometry and the hydrologic-state strata; SI12 outcome quality control and qualifiers; SI13 the history-dependent external arm; SI14 missingness and failure cases; SI15 reproduction hashes and environment parity; SI16 rights and data dictionary; SI19 development-period spatial analyses; SI20 the method, estimands, sealing, evidence-grade table, seal provenance and relocated objects of the Section 4.6 arms.

Figures S1–S3 give cohort geometry, temporal roles and model dataflow; S4–S8 expand Sections 4.3 and 4.5; S9 is a development-period conformal sensitivity not comparable with any held-out figure; S10 the architecture controls; S11 the matched spatial-transfer panels, relocated because they show a different model from the section that cites them and an effect the design cannot resolve.

References

Abatzoglou, J. T. (2013). Development of gridded surface meteorological data for

- ecological applications and modelling. *International Journal of Climatology*, 33, 121–131. (gridMET / METDATA) doi: 10.1002/joc.3413
- Arsenault, R., Brissette, F., & Martel, J.-L. (2018). The hazards of split-sample validation in hydrological model calibration. *Journal of Hydrology*, 566, 346–362. doi: 10.1016/j.jhydrol.2018.09.027
- Bakar Siddik, M. A., Turner, S., Gomez-Velez, J., Tanim, A., & Kao, S.-C. (2026). Advancing stream temperature prediction with a generalizable large-sample framework across CONUS river reaches. *Journal of Hydrology*, 666, 134780. doi: 10.1016/j.jhydrol.2025.134780
- Barclay, J. R., Topp, S. N., Koenig, L. E., Sleekman, M. J., & Appling, A. P. (2023). Train, inform, borrow, or combine? approaches to process-guided deep learning for groundwater-influenced stream temperature prediction. *Water Resources Research*, 59, e2023WR035327. doi: 10.1029/2023WR035327
- Caissie, D. (2006). The thermal regime of rivers: a review. *Freshwater Biology*, 51, 1389–1406. doi: 10.1111/j.1365-2427.2006.01597.x
- Chang, S. Y., Schwenk, J., & Solander, K. C. (2025). Deep learning advances Arctic river water temperature predictions. *Water Resources Research*, 61(6), e2024WR039053. doi: 10.1029/2024WR039053
- Corona, C. R., & Hogue, T. S. (2025). Machine learning in stream and river water temperature modeling: a review and metrics for evaluation. *Hydrology and Earth System Sciences*, 29, 2521–2549. doi: 10.5194/hess-29-2521-2025
- Feigl, M., Lebedzinski, K., Herrnegger, M., & Schulz, K. (2021). Machine-learning methods for stream water temperature prediction. *Hydrology and Earth System Sciences*, 25, 2951–2977. doi: 10.5194/hess-25-2951-2021
- Holm, S. (1979). A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2), 65–70. Retrieved from <https://www.jstor.org/stable/4615733>
- Jia, X., Zwart, J., Sadler, J., Appling, A., Oliver, S., Markstrom, S., ... Kumar, V. (2021). Physics-guided recurrent graph model for predicting flow and temperature in river networks. In *Proceedings of the 2021 siam international conference on data mining (sdm)* (pp. 612–620). SIAM. (Preprint: arXiv:2009.12575) doi: 10.1137/1.9781611976700.69
- Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., ... Liu, T.-Y. (2017). LightGBM: a highly efficient gradient boosting decision tree. In *Advances in neural information processing systems 30 (neurips 2017)* (pp. 3146–3154). Retrieved from <https://proceedings.neurips.cc/paper/2017/hash/6449f44a102fde848669bdd9eb6b76fa-Abstract.html>
- Kleměš, V. (1986). Operational testing of hydrological simulation models. *Hydrological Sciences Journal*, 31(1), 13–24. doi: 10.1080/02626668609491024
- Kratzert, F., Klotz, D., Herrnegger, M., Sampson, A. K., Hochreiter, S., & Nearing, G. S. (2019). Toward improved predictions in ungauged basins: Exploiting the power of machine learning. *Water Resources Research*, 55(12), 11344–11354. doi: 10.1029/2019WR026065
- Liu, J., Shen, C., O'Donncha, F., Song, Y., Zhi, W., Beck, H. E., ... Lawson, K.

- (2025). From RNNs to Transformers: benchmarking deep learning architectures for hydrologic prediction. *Hydrology and Earth System Sciences*, 29, 6811–6828. doi: 10.5194/hess-29-6811-2025
- Mohseni, O., Stefan, H. G., & Erickson, T. R. (1998). A nonlinear regression model for weekly stream temperatures. *Water Resources Research*, 34, 2685–2692. doi: 10.1029/98WR01877
- Nearing, G. S., Kratzert, F., Sampson, A. K., Pelissier, C. S., Klotz, D., Frame, J. M., ... Gupta, H. V. (2021). What role does hydrological science play in the age of machine learning? *Water Resources Research*, 57(3), e2020WR028091. doi: 10.1029/2020WR028091
- Padrón, R. S., Zappa, M., Bernhard, L., & Bogner, K. (2025). Extended-range forecasting of stream water temperature with deep-learning models. *Hydrology and Earth System Sciences*, 29(6), 1685–1702. doi: 10.5194/hess-29-1685-2025
- Philippus, D., Corona, C. R., & Hogue, T. S. (2026). Daily stream water temperature modeling and forecasting for ungaged watersheds at the CONUS scale. *Journal of Hydrology*, 675, 135620. doi: 10.1016/j.jhydrol.2026.135620
- Rahmani, F., Appling, A., Feng, D., Lawson, K., & Shen, C. (2023). Identifying structural priors in a hybrid differentiable model for stream water temperature modeling. *Water Resources Research*, 59, e2023WR034420. doi: 10.1029/2023WR034420
- Rahmani, F., Lawson, K., Ouyang, W., Appling, A., Oliver, S., & Shen, C. (2021). Exploring the exceptional performance of a deep learning stream temperature model and the value of streamflow data. *Environmental Research Letters*, 16, 024025. doi: 10.1088/1748-9326/abd501
- Sadler, J. M., Appling, A. P., Read, J. S., Oliver, S. K., Jia, X., Zwart, J. A., & Kumar, V. (2022). Multi-task deep learning of daily streamflow and water temperature. *Water Resources Research*, 58, e2021WR030138. doi: 10.1029/2021WR030138
- Seibert, J., Vis, M. J. P., Lewis, E., & van Meerveld, H. J. (2018). Upper and lower benchmarks in hydrological modelling. *Hydrological Processes*, 32(8), 1120–1125. doi: 10.1002/hyp.11476
- Thornton, M. M., Shrestha, R., Wei, Y., Thornton, P. E., & Kao, S.-C. (2022). *Daymet: daily surface weather data on a 1-km grid for North America, version 4 R1*. ORNL DAAC, Oak Ridge, Tennessee, USA. doi: 10.3334/ORNLDAAAC/2129
- Toffolon, M., & Piccolroaz, S. (2015). A hybrid model for river water temperature as a function of air temperature and discharge. *Environmental Research Letters*, 10, 114011. doi: 10.1088/1748-9326/10/11/114011
- Topp, S. N., Barclay, J., Diaz, J., Sun, A. Y., Jia, X., Lu, D., ... Appling, A. P. (2023). Stream temperature prediction in a shifting environment: explaining the influence of deep learning architecture. *Water Resources Research*, 59, e2022WR033880. doi: 10.1029/2022WR033880
- United States Geological Survey. (1994). *USGS water data for the nation*. National Water Information System (NWIS) daily values accessed via the data-

trieval Python client; parameter codes 00010, 00060, 00065 with statistic
code 00003. (Accessed 2026; <https://waterdata.usgs.gov/nwis>) doi:
10.5066/F7P55KJN

Zwart, J. A., Diaz, J., Hamshaw, S., Oliver, S., Ross, J. C., Sleckman, M., ...
White, E. (2023). Evaluating deep learning architecture and data assim-
ilation for improving water temperature forecasts at unmonitored locations.
Frontiers in Water, 5, 1184992. doi: 10.3389/frwa.2023.1184992

Zwart, J. A., Oliver, S. K., Watkins, W. D., Sadler, J. M., Appling, A. P., Corson-
Dosch, H. R., ... Read, J. S. (2023). Near-term forecasts of stream temper-
ature using deep learning and data assimilation in support of management
decisions. *JAWRA Journal of the American Water Resources Association*,
59(2), 317–337. doi: 10.1111/1752-1688.13093